Introduction

Reinforcement Learning

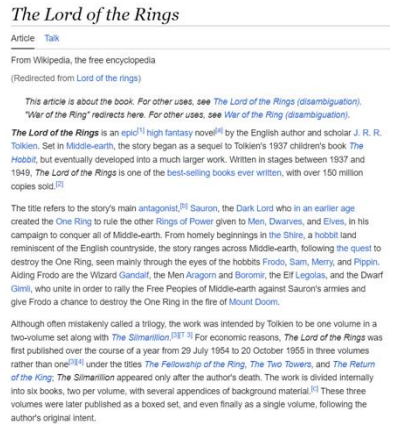
Main Areas of Artificial Intelligence (AI)

tabular data

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	MoSold	YrSold	SaleType	SaleCondition	SalePrice
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	2008	WD	Normal	208500
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	5	2007	WD	Normal	181500
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	9	2008	WD	Normal	223500
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	2006	WD	Abnorml	140000
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	12	2008	WD	Normal	250000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1455	1456	60	RL	62.0	7917	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	8	2007	WD	Normal	175000
1456	1457	20	RL	85.0	13175	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	MnPrv	NaN	0	2	2010	WD	Normal	210000
1457	1458	70	RL	66.0	9042	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	GdPrv	Shed	2500	5	2010	WD	Normal	266500
1458	1459	20	RL	68.0	9717	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	4	2010	WD	Normal	142125
1459	1460	20	RL	75.0	9937	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	6	2008	WD	Normal	147500

[1460 rows x 81 columns]

language models



The Lord of the Rings

Article Talk

From Wikipedia, the free encyclopedia

(Redirected from Lord of the rings)

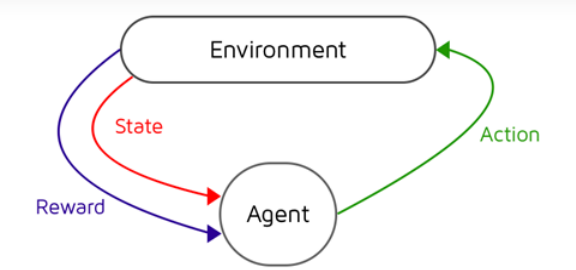
This article is about the book. For other uses, see The Lord of the Rings (disambiguation). "War of the Ring" redirects here. For other uses, see War of the Ring (disambiguation).

The Lord of the Rings is an epic[1] high fantasy novel[4] by the English author and scholar J. R. R. Tolkien. Set in Middle-earth, the story began as a sequel to Tolkien's 1937 children's book *The Hobbit*, but eventually developed into a much larger work. Written in stages between 1937 and 1949, *The Lord of the Rings* is one of the best-selling books ever written, with over 150 million copies sold.[2]

The title refers to the story's main antagonist,[5] Sauron, the Dark Lord who in an earlier age created the One Ring to rule the other Rings of Power given to Men, Dwarves, and Elves, in his campaign to conquer all of Middle-earth. From homely beginnings in the Shire, a hobbit land reminiscent of the English countryside, the story ranges across Middle-earth, following the quest to destroy the One Ring, seen mainly through the eyes of the hobbits Frodo, Sam, Merry, and Pippin. Aiding Frodo are the Wizard Gandalf, the Men Aragorn and Boromir, the Elf Legolas, and the Dwarf Gimli, who unite in order to rally the Free Peoples of Middle-earth against Sauron's armies and give Frodo a chance to destroy the One Ring in the fire of Mount Doom.

Although often mistakenly called a trilogy, the work was intended by Tolkien to be one volume in a two-volume set along with *The Silmarillion*.[3][T 3] For economic reasons, *The Lord of the Rings* was first published over the course of a year from 29 July 1954 to 20 October 1955 in three volumes rather than one[3][4] under the titles *The Fellowship of the Ring*, *The Two Towers*, and *The Return of the King*; *The Silmarillion* appeared only after the author's death. The work is divided internally into six books, two per volume, with several appendices of background material.[6] These three volumes were later published as a boxed set, and even finally as a single volume, following the author's original intent.

control



computer vision

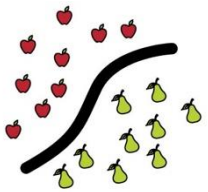
all empowered by one key component:
learning from data → Machine Learning (ML)

MACHINE LEARNING

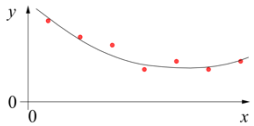
training target available (labeled or past data)

SUPERVISED

CLASSIFICATION



REGRESSION



data not labeled in any way

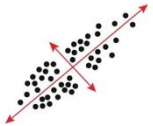
UNSUPERVISED

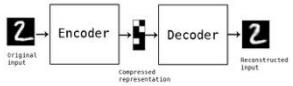
CLUSTERING

SELF-SUPERVISED

ASSOCIATION

DIMENSIONALITY REDUCTION





no supervision, but goal-based interaction with environment

REINFORCEMENT LEARNING

LEARN STATE OR ACTION VALUES

LEARN POLICY DIRECTLY

learning by teacher (high-dimensional curve fitting)

learning by observation (pattern recognition)

learning by trial-and-error (sequential decision making)

Course Schedule

1. Foundations

- Motivation: control theory refresher, Model Predictive Control, control vs learning
- Mathematical Setup: Markov Decision Processes, rewards, policies, state-value & action-value functions, Bellman equations
- Planning: policy evaluation & improvement, value iteration, policy iteration, dynamic programming
- Model-Free Learning: Monte Carlo methods, Temporal-Difference learning, SARSA, Q-learning

2. Deep RL

- Function Approximation: limitations of tabular methods, machine learning, deep learning
- Value-Based Deep RL: Deep-Q Networks, replay buffers, target networks, distribution shift in RL, the deadly triad, stabilization principles
- Policy Gradient Methods: direct policy search, policy gradient theorem, log-likelihood trick, REINFORCE, baselines & variance reduction
- Actor–Critic Methods: Proximal Policy Optimization, Soft Actor-Critic

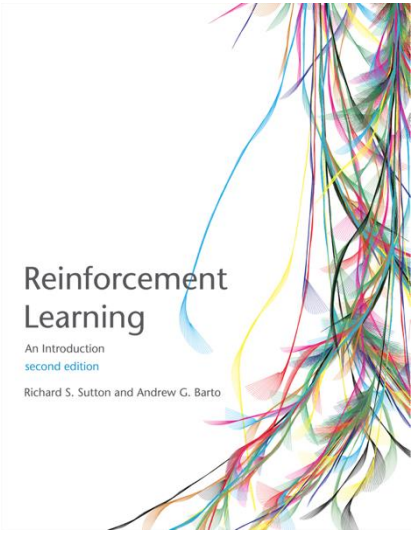
3. Advanced & Emerging Methods

- Learning with Models and Fixed Data: model-based RL, offline RL
- Learning from Demonstrations and Humans: behavior cloning, imitation learning, RL from Human Feedback
- RL as Sequence and Generative Modeling: trajectory modeling, Decision Transformer, diffusion policies
- Scaling RL to Real-World Robotics: embodied RL, sim-to-real gap, generalist robot policies, Vision–Language–Action models

Literature

Reinforcement Learning: An Introduction

classic textbook
covering the foundations



Reinforcement Learning: An Overview

includes Deep RL and
some advanced topics

both freely available as pdf

Used Programming Language: Python

- good compromise between rapid prototyping and production
- vast ecosystem
- very popular for data processing and ML: scientific Python stack

NumPy

matplotlib

PyTorch

pandas

Gymnasium

scikit learn

Programming Environments

locally, best use

- a virtual environment to flexibly install packages (e.g., venv)
- an IDE of your choice (e.g., VS Code)
- both plain Python files or Jupyter notebooks are fine

but cloud-based environments also fine (e.g., Google Colab)